

FAIR COMPLIANCE ASSESSMENT

Tier 3 :: Poor Data Stewardship

At a glance: FAIR compliance assessment tools evaluate whether a research dataset or repository meets the FAIR principles — Findable, Accessible, Interoperable, and Reusable — published by Wilkinson et al. in 2016. Unlike the other tools in this guide, FAIR assessment does not detect misconduct in published papers. Instead, it measures the structural quality of data stewardship: whether data can be found by a machine, retrieved without restriction, understood in another context, and legally reused by other researchers. Poor FAIR compliance is a field-level indicator of systemic data loss and reproducibility risk.

1. Overview

The reproducibility crisis has two distinct dimensions. The detection tools in this guide address the first: statistical and methodological failures in published research. The FAIR principles address the second: the structural failure of research data infrastructure. Even if every paper in a literature were methodologically perfect, a field in which datasets are inaccessible, poorly described, or stored in proprietary formats cannot be reproduced, reanalyzed, or built upon. Data that cannot be found or understood is, for scientific purposes, data that does not exist.

The FAIR Guiding Principles, published in *Scientific Data* by Wilkinson and colleagues in 2016, articulate the properties that data and their associated metadata must have to be genuinely reusable. The principles were designed to be implemented by both humans and machines — recognizing that at the scale of modern science, manual assessment of data quality is not feasible. FAIR compliance assessment tools operationalize the principles into scorable, automatable criteria that can be applied to individual datasets, repository-wide collections, or entire research programs.

FAIR assessment sits at Tier 3 in this framework — it is a field-level tool rather than a paper-level or study-level one. A single dataset failing a FAIR check is a management problem. A systematic pattern of low FAIR compliance across a research domain is a structural integrity problem: it means the field cannot audit its own evidence base, replication is practically impossible even for well-intentioned researchers, and the published literature is built on a foundation that cannot be independently verified.

2. The FAIR Principles in Detail

Each of the four principles — Findable, Accessible, Interoperable, Reusable — is operationalized into a set of sub-principles, each of which can be scored independently. The sub-principles are numbered F1–F4, A1–A1.2, I1–I3, and R1–R1.3 in the original Wilkinson et al. paper. FAIR assessment tools map each sub-principle to a testable criterion.

	Principle	What it requires	Why it matters for detection
F	Findable	Persistent identifier (DOI, ARK, PURL) assigned to both data and metadata. Rich metadata describing the data is indexed in a searchable resource. Metadata explicitly includes the identifier for the data it describes.	Datasets that cannot be found by machines cannot be automatically aggregated for meta-analysis, replication matching, or bibliometric analysis. A literature of unfindable data cannot be systematically reviewed.
A	Accessible	Data and metadata are retrievable via a standardized, open, and free protocol (e.g., HTTP, FTP). Authentication and authorization are permitted but must use open, implementable standards. Metadata remains accessible even if the data itself is no longer available.	Gated or expired data links make computational reproducibility impossible. Accessible metadata-after-data-deletion preserves at least the record of what existed, enabling detection of selective data retention.
I	Interoperable	Data use a formal, accessible, shared, and broadly applicable language for knowledge representation. Metadata use vocabularies that follow FAIR principles. Metadata include qualified references to other data.	Interoperable data can be automatically combined across studies — enabling automated p-curve construction, replication matching, and cross-study consistency checking without manual harmonization.
R	Reusable	Metadata include a plurality of accurate and relevant attributes. Data are released with a clear and accessible data usage license. Metadata meet	Without a clear license, data cannot legally be reused for reanalysis. Without provenance, the chain of custody from raw collection to

		domain-relevant community standards. Data and metadata are associated with their provenance.	published summary statistic cannot be audited.
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3. FAIR Assessment Tools

Several automated and semi-automated tools have been developed to score datasets against the FAIR sub-principles. They differ in their scope, the level of automation they achieve, and the communities for which they were designed.

F-UJI (Automated FAIR Data Assessment Tool)

F-UJI, developed by Devaraju and Huber (2021) at the German National Library of Science and Technology, is a fully automated web service that accepts a dataset's persistent identifier or landing page URL and returns a scored assessment across 17 FAIR sub-principles. It uses a combination of content negotiation, metadata harvesting, and vocabulary lookup to assess each criterion without human intervention.

F-UJI is the most widely used automated FAIR assessment tool and is recommended by the European Open Science Cloud (EOSC) for repository compliance monitoring. It returns a structured JSON result with pass/fail scores for each sub-principle, a maturity level score (0–4) for each FAIR category, and an overall FAIR score. The tool is accessible via a REST API, making it suitable for batch assessment of large repository collections.

Input	A persistent identifier (DOI, Handle, ARK) or the URL of the dataset's landing page.
Output	JSON result with sub-principle scores, maturity levels (0–4) per FAIR category, and an overall compliance summary. A web UI at f-ujl.net provides a human-readable report.
Scope	Assesses the metadata and structural properties of the dataset landing page. Does not evaluate the scientific quality or completeness of the data itself.
Automation	Fully automated. No human judgment required for standard assessments. Can be called programmatically via API for batch evaluation of repository holdings.
Limitations	Relies on publicly accessible metadata. Datasets in repositories that use non-standard metadata formats or that restrict metadata access may score poorly even when FAIR-compliant in practice.

FAIRshake

FAIRshake, developed by the Ma'ayan Laboratory at the Icahn School of Medicine at Mount Sinai, takes a community-driven approach. Rather than a single automated scoring algorithm, FAIRshake provides a framework for defining FAIR rubrics tailored to specific communities — biomedical databases, computational tools, clinical datasets — and allows both human reviewers and automated scripts to apply those rubrics.

FAIRshake was used extensively in the NIH Data Commons pilot program and has been applied to assess compliance across large biomedical data repositories including GEO, dbGaP, and the Metabolomics Workbench. Its community-specific rubrics make it more sensitive to domain norms than generic tools, at the cost of requiring rubric development effort before application.

Input	Dataset URL, identifier, or repository API endpoint. Community-specific rubric must be selected or defined before scoring.
Output	Per-criterion scores, visual FAIR insignia (a standardized badge system), and aggregate compliance statistics across a collection.
Scope	Can assess individual datasets, tools, or entire repository collections. Particularly well-suited for community-specific compliance monitoring.
Automation	Partially automated. Core metadata checks are automated; some rubric criteria require human judgment. Batch API access available.

Limitations	Requires community-specific rubric development. Results are not directly comparable across communities using different rubrics.
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FAIR Evaluator

The FAIR Evaluator, maintained by the GO FAIR initiative and accessible at evaluator.fairdata.org, focuses on machine-actionable FAIR — specifically testing whether metadata is structured in a way that a software agent could discover, retrieve, and interpret the dataset without human assistance. It applies a set of FAIR Maturity Indicators, each of which is tested by a specific automated compliance test.

The FAIR Evaluator is particularly relevant for precision medicine and biomedical research, where the goal is not just that a human researcher can find and use data, but that computational pipelines can automatically integrate data across studies. It places heavier weight on semantic interoperability — whether metadata uses controlled vocabularies and ontologies that machines can reason about — than general-purpose tools like F-UJI.

4. Example Application: NIH-Funded Research Repositories

Gabelica et al. (2022, *Journal of Clinical Epidemiology*) audited 3,556 biomedical papers that included a data availability statement claiming the data were available upon reasonable request. They then systematically attempted to obtain the data by contacting authors. The results were stark: 79% of authors either did not respond, refused to share, or shared data that was incomplete or unusable. Less than 7% of the claimed-available datasets were actually accessible.

A parallel assessment using F-UJI on the small fraction of datasets that were deposited in repositories found that even deposited datasets frequently failed key FAIR sub-principles: only 32% had persistent identifiers, 41% had machine-readable metadata, and fewer than 20% used a recognized vocabulary for describing their variables. The datasets existed but were not practically reusable.

This study illustrates the two-layer failure mode that FAIR assessment reveals: the first layer is non-deposition (data never shared despite stated availability), detectable through audit. The second layer is inadequate FAIR compliance in deposited data (data technically available but practically unusable), detectable through automated assessment tools. Both layers contribute to the broader data stewardship failure that FAIR assessment is designed to diagnose.

A second application comes from the NIH data sharing mandate implemented in January 2023, which requires that all NIH-funded research generating scientific data include a data management and sharing plan. The NIH Office of Data Science Strategy uses F-UJI-like assessment frameworks to evaluate compliance with the mandate, identifying which funded projects are producing FAIR-compliant data deposits and which are meeting the letter but not the spirit of the requirement.

5. Requirements & Applicability

FAIR assessment tools have very different requirements from the statistical detection tools in this guide. They evaluate infrastructure and metadata quality rather than statistical reporting, and they operate on data repositories rather than published papers.

A dataset landing page or persistent identifier	F-UJI and the FAIR Evaluator require a URL or DOI pointing to the dataset's landing page or metadata record. Datasets without any online presence cannot be assessed. This is itself a FAIR failure (sub-principle F1: data must have a globally unique and persistent identifier).
Publicly accessible metadata	Automated tools assess metadata that is publicly retrievable via standard web protocols. Datasets in restricted repositories, or whose metadata requires authentication to access, will score lower than their actual compliance warrants. For restricted data, manual assessment using a rubric is the alternative.
No raw data access required	FAIR assessment tools evaluate metadata and structural properties, not the data content itself. The data files need not be downloaded. This makes assessment scalable to large repository collections.

Repository or field scope definition	FAIR assessment is most informative when applied systematically to a defined collection: all datasets deposited in support of papers published in a specific journal, all datasets funded by a specific grant program, or all data in a disciplinary repository. Single-dataset assessments are less interpretable without a baseline comparison.
Domain context for interpretation	FAIR sub-principle scores require domain knowledge to interpret. A dataset in a field with weak community metadata standards may score lower than a dataset in a field with mature ontologies, even if the first is more carefully curated. Comparison across fields without accounting for infrastructure maturity can be misleading.

6. FAIR Assessment as a Research Integrity Signal

FAIR compliance is typically framed as a data management goal rather than an integrity tool. But systematic FAIR assessment reveals patterns that are directly relevant to research integrity at the field level. Several specific signals are worth highlighting for graduate seminars and meta-research contexts.

Data availability statement audits

The gap between stated and actual data availability — documented by Gabelica et al. (2022) and Vines et al. (2014) — is itself a form of misrepresentation. Papers that claim data are available when they are not are reporting false information in the methods or transparency sections. Systematic audit of data availability statements, combined with F-UJI assessment of actually deposited datasets, provides a field-level measure of transparency integrity.

Selective data retention

The FAIR principle that metadata should remain accessible even after data deletion (sub-principle A2) exists precisely because selective deletion of data is a form of evidence manipulation. When a dataset is deleted after publication — particularly when the deletion occurs following concerns about the underlying research — the metadata record provides at least a forensic trace of what existed. Repositories that implement A2 correctly preserve this audit trail; those that do not enable clean evidence destruction.

Provenance as a chain of custody

The FAIR reusability principle requires that data be associated with their provenance — the record of how data were collected, processed, and transformed from raw measurements to the form in which they are deposited. Provenance metadata is the data equivalent of a laboratory notebook: it documents the chain of custody from observation to archive. Datasets with complete provenance metadata can be audited for consistency with the methods section of the associated paper; datasets without provenance cannot.

Compliance rates as field-level integrity indicators

When FAIR assessment is applied systematically across a field's data deposits, the resulting compliance statistics become an integrity indicator. A field in which fewer than 20% of datasets have persistent identifiers, or in which 79% of stated-available datasets are actually inaccessible, has a structural integrity problem that no amount of statistical sophistication in individual papers can compensate for. These field-level rates are now being tracked by funders, publishers, and research assessment frameworks as components of responsible research practice evaluation.

7. Limitations

- FAIR compliance measures infrastructure quality, not scientific quality. A meticulously described dataset of fabricated measurements is FAIR-compliant. High FAIR scores are a necessary but not sufficient condition for data integrity.
- Automated assessment tools evaluate metadata properties, not the accuracy or completeness of the metadata itself. A dataset can have a DOI, use a recognized vocabulary, and carry an open license, while containing inaccurate variable descriptions or omitting crucial methodological details. FAIR assessment cannot catch lies told in metadata.

- FAIR sub-principle scores are not fully standardized across tools. F-UJI, FAIRshake, and the FAIR Evaluator use different scoring rules for some criteria, making cross-tool comparisons unreliable. When reporting FAIR compliance results, always specify which tool and which version was used.
- Many disciplines lack mature community metadata standards — the vocabulary and ontology infrastructure that FAIR interoperability and reusability require. In such fields, even well-intentioned data depositors will score poorly on these sub-principles through no fault of their own. FAIR assessment in immature data ecosystems measures the infrastructure gap as much as individual compliance.
- FAIR does not address data quality, completeness, or representativeness. A dataset can be perfectly FAIR-compliant and still be a biased sample, collected with a flawed instrument, or analyzed with inappropriate methods. The FAIR framework is about access and reusability, not scientific validity.

8. How to Apply

Assessing an individual dataset (F-UJI)

Navigate to f-uji.net and enter the dataset's DOI or landing page URL. The tool returns a scored report within seconds. Review the sub-principle scores, noting which categories (F, A, I, R) show low maturity. The most common failures in published research datasets are: no persistent identifier (F1), no structured machine-readable metadata (F2), restrictive or absent license (R1.1), and no provenance information (R1.2).

Batch assessment of a repository collection

Use the F-UJI REST API (documented at f-uji.net/api) to programmatically assess a collection of datasets. Collect DOIs from a bibliometric database or journal data availability statements, submit each to the API, and aggregate the JSON results. The resulting dataset of sub-principle scores can then be analyzed to produce field-level compliance statistics comparable to those reported by Gabelica et al. (2022).

Auditing data availability statements

For a systematic audit: (1) collect all data availability statements from a sample of papers in the target journal or field; (2) classify each statement (data in repository / available on request / not available / no statement); (3) for statements claiming repository deposit, extract the DOI or URL and run F-UJI; (4) for statements claiming availability on request, contact authors following the protocol used by Gabelica et al. Document response rates, refusal rates, and the FAIR compliance of actually obtained datasets. The gap between stated and actual availability, and between deposited and FAIR-compliant datasets, are the two primary integrity signals.

9. Related Tools

DataCite	Global DOI registration agency for research data. DataCite DOIs are the most widely used persistent identifiers for datasets (FAIR sub-principle F1). The DataCite Metadata Schema specifies the structured metadata required for registration, covering many F and R sub-principles automatically for registered datasets.
ORCID	Persistent researcher identifier that links publications, datasets, and other outputs to a verified scholarly identity. Contributes to FAIR provenance (R1.2) by unambiguously attributing data to specific researchers across institutions and career changes.
Zenodo / Figshare / Dryad	Open data repositories that implement FAIR principles structurally: automatic DOI assignment (F1), open HTTP access (A1), Dublin Core and DataCite metadata standards (I1, R1), and Creative Commons licensing (R1.1). Depositing in these repositories confers substantial FAIR compliance without additional metadata effort.
Research Object Crate (RO-Crate)	A lightweight standard for packaging research data with structured metadata using Schema.org vocabularies. RO-Crate directly implements FAIR interoperability (I1–I3) and provenance (R1.2) requirements. Increasingly adopted in workflow management systems and computational reproducibility pipelines.
Registered Reports	Preregistration format that commits to data sharing before data collection. Journals offering Registered Reports typically require FAIR-compliant data deposits as a condition of publication — creating a structural link between study pre-registration and data accessibility that addresses the data availability gap.

Key Citations

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